Supplementary Material for PrivECE: Enhanced Conditional Estimation on Key-Value Data with Local Privacy Guarantees

Anonymous Author(s)

1 Additional Theoretical Analysis

1.1 Separation under Value-Domain Extension

Proposition 1 (Separation induced by domain extension). *Let $|\mathcal{V}_{\text{ori}}| = d$ and $|\mathcal{V}_{\text{ext}}| = \beta d$ for some $\beta > 0$. Let the extended true distribution be $\boldsymbol{\pi}^* = [\boldsymbol{\pi}, \mathbf{0}]$. Then for*GRR *defined on $\mathcal{V}_{\text{ori}} \cup \mathcal{V}_{\text{ext}}$, the resulting mixture satisfies*

$$\Delta_{\text{ext}} = \text{TV}(G\boldsymbol{\pi}^*, \text{Unif}) > 0.$$

Let $D = (1+\beta)d$ be the extended output domain size. Under D-ary GRR with privacy budget ε, the mechanism satisfies: for any true input value x, the reported value equals x with probability

$$p = \frac{e^\varepsilon}{e^\varepsilon + D - 1},$$

and equals any other value with probability

$$q = \frac{1}{e^\varepsilon + D - 1}.$$

Consider the extended true distribution $\pi^* = [\pi, \mathbf{0}]$ supported only on $\mathcal{V}_{\text{ori}}$. For any $i \in \mathcal{V}_{\text{ori}}$, the probability of observing output i is

$$(G_D\pi^*)_i = \sum_{x \in \mathcal{V}_{\text{ori}}} \pi_x \Pr[Y = i \mid X = x] = q + (p - q)\pi_i.$$

For any $j \in \mathcal{V}_{\text{ext}}$, since X never takes values in $\mathcal{V}_{\text{ext}}$, the output j can only occur as an "other" value under GRR, hence

$$(G_D\pi^*)_j = \sum_{x \in \mathcal{V}_{\text{ori}}} \pi_x \Pr[Y = j \mid X = x] = \sum_{x \in \mathcal{V}_{\text{ori}}} \pi_x\, q = q.$$

Let u_{ext} denote the uniform distribution on the extended domain, i.e., $(u_{\text{ext}})_v = 1/D$ for all v. Then by definition of total variation distance,

$$\Delta_{\text{ext}} = \frac{1}{2} \sum_{v \in \mathcal{V}_{\text{ori}} \cup \mathcal{V}_{\text{ext}}} \left| (G_D\pi^*)_v - \frac{1}{D} \right|.$$

Dropping the nonnegative contribution from the original-domain terms yields a lower bound:

$$\Delta_{\text{ext}} \geq \frac{1}{2} \sum_{j \in \mathcal{V}_{\text{ext}}} \left| q - \frac{1}{D} \right| = \frac{\beta d}{2} \left| q - \frac{1}{D} \right|.$$

Since $e^\varepsilon > 1$, we have

$$q = \frac{1}{e^\varepsilon + D - 1} < \frac{1}{D},$$

and therefore $\left|q - \frac{1}{D}\right| = \frac{1}{D} - q > 0$, implying $\Delta_{\text{ext}} > 0$ for any $\beta > 0$.

Finally, if $\Delta_{\text{ori}} < \frac{\beta d}{2}\left(\frac{1}{D} - q\right)$, then the above lower bound already exceeds Δ_{ori}, hence $\Delta_{\text{ext}} > \Delta_{\text{ori}}$. In particular, when π is uniform on $\mathcal{V}_{\text{ori}}$, one has $G_d\pi = u_d$ and thus $\Delta_{\text{ori}} = 0$, while $\Delta_{\text{ext}} > 0$.

1.2 Supporting Accuracy Analysis

Throughout, fix the Phase-I outputs, including $\mathcal{K}_{\text{pop}}$ and s. Users in $\mathcal{U}_2$ are i.i.d., and public sub-domains are sampled independently of users' data.

Sampling. The expected number of reports covering key k is $n_2 s/d_{\text{pop}}$, and the true-value fraction among these reports is $\rho_k = f_k a_k$. Theorem 2 uses the actual number $N_k > 0$.

Proof of Theorem 2. For GRR, direct calculation gives

$$A^\top A = \lambda_M^2 \left(I_L - \frac{\mathbf{1}\mathbf{1}^\top}{M} \right),$$

$$(A^\top A)^{-1} = \lambda_M^{-2} \left(I_L + \frac{\mathbf{1}\mathbf{1}^\top}{B} \right).$$

Thus, when $B > 0$, the report distribution uniquely determines the true-value weights and the dummy proportion.

Every report probability is at most $p_\varepsilon = e^\varepsilon/(e^\varepsilon + M - 1)$. Using $\mathcal{I}_k = A^\top \operatorname{diag}(\mathbf{h}_k)^{-1} A$ and $Mp_\varepsilon \leq e$ gives

$$\mathcal{I}_k^{-1} \preceq p_\varepsilon (A^\top A)^{-1} \preceq \frac{e}{B\lambda_M^2} I_L.$$

Under the theorem's assumptions, the leading covariance of $\widehat{\boldsymbol{\pi}}_k$ is $\mathcal{I}_k^{-1}/N_k$. Let $\mathbf{g}_k = \boldsymbol{\alpha} - \mu_k \mathbf{1}$. A first-order Taylor expansion gives

$$\text{AVar}(\widehat{f}_k) = \frac{\widehat{\xi}_s^2}{N_k} \mathbf{1}^\top \mathcal{I}_k^{-1} \mathbf{1},$$

$$\text{AMSE}(\widehat{\mu}_k) = \frac{1}{N_k \rho_k^2} \mathbf{g}_k^\top \mathcal{I}_k^{-1} \mathbf{g}_k.$$

Since $\|\mathbf{1}\|_2^2 = L$ and $\|\mathbf{g}_k\|_2^2 \leq 4L$, the preceding matrix bound proves Equation (22).

Expansion and remaining error. For fixed L and ε, the expansion factor $L/(B\lambda_M^2)$ eventually increases with B. Thus, adding too many categories increases the error bounds. For SW, the corresponding analysis must use its own perturbation probabilities.

2 Dataset Details

We empirically evaluate the proposed key value mechanisms on both synthetic and real-world datasets. The real-world datasets are selected following prior work, and the Taxi dataset is used to evaluate the setting where the value in key–value data is two-dimensional.

- *Synthetic Datasets*. Two synthetic datasets are generated according to gaussian distribution and power-law distribution respectively, which are marked as GAUSS and PLAW. Specifically, the number of keys is fixed to 100 for Gauss, and 300 for PLAW, and we assume the availability of prior knowledge on ℓ forPCKV. Each user possesses approximately the same number of key–value pairs. Moreover, both the key frequencies and the mean values associated with each key are generated according to either a Gaussian or a power-law distribution.
- *MovieLens*. This dataset is publicly available and widely used for evaluating recommendation systems. We use the MovieLens-32M dataset, which contains approximately 32 million rating records from about 0.2 million users. Each record corresponds to a user

rating a movie, where the movie title serves as the key and the rating value serves as the numerical value.

- *Amazon*. This dataset has been used in PCKV. It comprises approximately 2 million ratings sourced from unique users for a total of 145,790 distinct keys.
- *Taxi*. This dataset is derived from the New York City Taxi trip records, which are widely used for urban data analysis. Each trip is modeled as a key-value record, where the trip duration category serves as the key and the (pickup region, drop-off region) pair serves as the value. Trip durations are discretized into six categories, and we treat the pickup drop-off region index as a two dimension value. The value estimation task is to analyze the correlation between pickup and drop-off regions across different trip duration categories.

Further detials can be seen in Table 1. According to the average number of users per key (i.e., $\frac{\#\text{Users}}{\#\text{keys}}$), these datasets can be categorized into three groups - dense population (GAUSS, PLAW and Taxi), medium population (Amazon) and sparse population (Movie).

Table 1: Statistics of Datasets

Dataset	#Users	#Keys	#Records(per user)	Avg.of f_k	Std.of f_k
GAUSS	10^6	100	11.23	0.1123	0.177
PLAW	10^6	300	4.43	0.0148	0.011
MovieLens[1]	200948	14395	159.2	0.0193	0.025
Amazon[2]	1,210,271	145,790	1.74	$6.71e{-6}$	$8.88e{-10}$
Taxi[3]	1,458,644	6	1	0.167	0.031

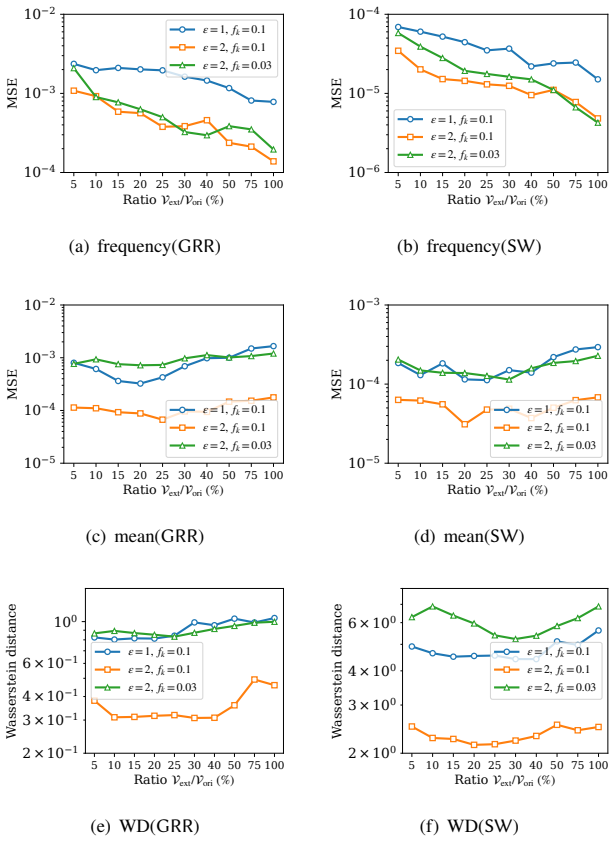
3 Additional Experiments

In this part, we also evaluate the density of value by Wasserstein distance (WD), which is the cost of moving the probability mass (or density) from one distribution to another distribution. The ℓ_1 loss of Wasserstein distance is

$$\mathrm{WD}(\mathbf{w}, \hat{\mathbf{w}}) = \sum_{\beta \in \mathcal{B}} |\mathbf{P}(\mathbf{w}, \beta) - \mathbf{P}(\hat{\mathbf{w}}, \beta)|.$$

Influence of value domain extension size. Figure 1 illustrates the performance of frequency estimation, conditional mean estimation, and distribution estimation (measured by the Wasserstein distance) as the size of the extended value domain $\tilde{\mathcal{V}}_{\text{ext}}$ varies on the synthetic dataset. In particular, the values are drawn from a Gaussian distribution with mean 0 and variance 2, and the value range is truncated at two standard deviations to ensure that the original value distribution $\boldsymbol{\pi}$ is dense and free of sparsity effects. The x-axis represents the ratio $|\tilde{\mathcal{V}}_{\text{ext}}|/|\tilde{\mathcal{V}}_{\text{ori}}|$. The size of $\mathcal{V}_{\text{ori}}$ is set to 20 for GRR-based perturbation and 256 for SW-based perturbation. For example, when the ratio is 25%, the extended domain size is 5 under GRR and 64 under SW. We evaluate three representative conditional settings (e.g., $f_k = 0.1$, $\varepsilon = 1$) to examine the effect of domain expansion.

As $\tilde{\mathcal{V}}_{\text{ext}}/\tilde{\mathcal{V}}_{\text{ori}}$ increases, the accuracy of frequency estimation consistently improves, since a larger extended domain enhances the identifiability of mixture components in the EM model. In contrast, for value-related tasks such as conditional mean and distribution estimation, the error increases once the extended domain exceeds a certain threshold, due to amplified perturbation noise in a larger value space. Overall, these results reveal a tradeoff between frequency estimation and value estimation when selecting the size of the extended value domain. Based on Figure 1, we suggest choosing $\tilde{\mathcal{V}}_{\text{ext}}/\tilde{\mathcal{V}}_{\text{ori}}$ in the range of 20% to 50%.



(a) frequency(GRR) (b) frequency(SW)

(c) mean(GRR) (d) mean(SW)

(e) WD(GRR) (f) WD(SW)

Figure 1: Results of parameter selection for $\mathcal{V}_{\text{ext}}$ under different tasks, using SW and GRR as value perturbation mechanisms.

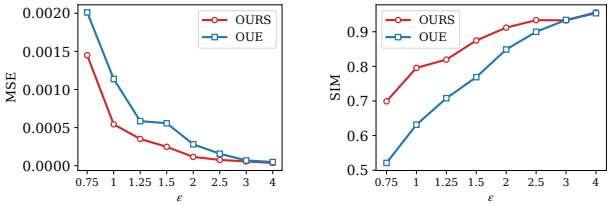
Two-Dimensional Values in the KV Setting. This dataset is derived from the New York City Taxi trip records, which are widely used in urban data analysis. Each trip is modeled as a key-value record, where the trip duration category serves as the key and the (pickup region, drop-off region) pair serves as a two-dimensional value. Trip durations are discretized into six categories.

To protect the two-dimensional values, we apply the LVP with multi-SW mechanism [2] serving as the value perturbation method. The value estimation task aims to analyze the correlation between pickup and drop-off regions under different trip duration categories. Specifically, we evaluate (1) the Pearson correlation between pickup and drop-off regions conditioned on each trip duration category, and (2) the frequency distribution of different trip duration categories.

We compare our method with OUE [1] as the baseline, where the trip duration, pickup region, and drop-off region are combined into a single categorical attribute, since most existing KV correlation baselines do not support distribution estimation for multi-dimensional values. We also exclude PrivKVM* from the comparison because its

estimation error becomes excessively large when the value domain is large.

As shown in Figure 2, our method consistently outperforms OUE in correlation estimation accuracy across different travel-duration categories. The vertical axis in Figure 2(b) reports Pearson similarity, with higher values indicating better performance.



(a) Taxi(frequency) (b) Taxi(pearsonr similarity)

Figure 2: Results on the NYC Taxi dataset.

4 Notations

References

[1] T. Wang, J. Blocki, N. Li, and S. Jha. Locally differentially private protocols for frequency estimation. pages 729–745, 2017.

[2] Y. Ye, M. Zhang, and D. Feng. Collecting spatial data under local differential privacy. In *2021 17th International Conference on Mobility, Sensing and Networking (MSN)*, pages 120–127, 2021.

Table 2: Main notation used in PrivECE.

Symbol	Description
$\mathcal{K}_{\text{pop}}$	Popular-key set selected from the Phase-I frequency estimates and used for Phase-II pair selection.
$\mathcal{V}^+$	Extended input value domain: the disjoint union of the valid domain $\mathcal{V}_{\text{ori}}$ and the added domain $\mathcal{V}_{\text{ext}}$.
m_u	Number of popular keys held by user u: $m_u = \lvert I_u \cap \mathcal{K}_{\text{pop}} \rvert$.
R_u	Publicly sampled size-s subset of $\mathcal{K}_{\text{pop}}$ indexing user u's Phase-II report.
$\widehat{H}(m)$	Privately estimated probability mass function of the retained-key count m_u.
$c_s(m)$	Probability that a size-s public sub-domain intersects a user's set of m popular keys.
$\widehat{C}(s)$	Estimated population coverage: $\widehat{C}(s) = \sum_m \widehat{H}(m) c_s(m)$.
$\omega_s(m)$	Probability of selecting a particular held key conditional on its inclusion in the public sub-domain: $\omega_s(m) = d_{\text{pop}} c_s(m)/(sm)$ for $m > 0$.
N_k	Number of Phase-II reports containing coordinate k: $N_k = \lvert\{u \in \mathcal{U}_2 : k \in R_u\}\rvert$.
$\pi_{k,j}$	Unnormalized mixture weight of valid true-value component j among reports covering key k.
ρ_k	Total selected true-value mass among reports covering k: $\rho_k = \sum_{j \in \mathcal{I}_{\text{ori}}} \pi_{k,j} = 1 - f_{\text{dum},k}$.
a_k	Mean conditional selection probability for holders of key k: $a_k = \mathbb{E}[\omega_s(m_u) \mid k \in I_u]$, so $\rho_k = f_k a_k$.
$\widehat{\xi}_s$	Population-level sampling correction factor estimated from $\widehat{H}$ and $\widehat{C}(s)$; generally an approximation to a_k^{-1}.
$\mathcal{I}_k$	Per-observation Fisher information matrix for the valid true-value mixture weights: $\mathcal{I}_k = A^\top \operatorname{diag}(\boldsymbol{h}_k)^{-1} A$.
λ_M	GRR probability contrast on an extended domain of size M: $\lambda_M = (e^\varepsilon - 1)/(e^\varepsilon + M - 1)$.